

Topological Segmentation Error and Boundary-Condition Tuning in Computed Coronary FFR: A Controlled In Silico Study

Fadillah Yamin, Mohd Azan Mohammed Sapardi, Ming Kwang Tan, Xin Wang, and Mohd-Zulhilmi Paiz Ismadi

Abstract—Fractional flow reserve, the pressure ratio that decides whether a coronary narrowing is treated, can be computed from computed tomography angiography. The computation depends on the segmented artery and on outlet boundary conditions derived from it, often tuned to match measured perfusion. Whether tuning can hide a segmentation error that changes the treatment decision at 0.80 has not been tested. We inserted 150 stenoses into 108 coronary trees, applied four segmentation error types (missed side branch, vessel break, longer lesion, narrowed taper), and recomputed fractional flow reserve with a reduced-order model under three boundary-condition protocols: fixed, re-derived from the corrupted tree, and tuned to the clean tree's territory perfusion, in two microvascular bed structures. One case was solved in three dimensions. With fixed boundary conditions, the branching (topological) errors changed the decision in 32–33% of models, against 6–10% for vessel-size (caliber) errors of inter-observer magnitude, close to the variability of repeat invasive measurement. Re-derived or tuned boundary conditions reduced topological decision changes to 5–20%. Tuning, however, removed the perfusion mismatch that marked the error: 8–19% of tuned models with a topological error passed a perfusion check stricter than measurement repeatability while their fractional flow reserve was wrong by more than 0.05. In three dimensions, prescribed territory flows reduced a missed-branch shift of 0.074 to below 0.001. A perfusion-matched coronary model can thus pass validation with a material error in its fractional flow reserve. Detecting it requires a check of the segmentation's branching structure and of the perfusion mismatch before tuning.

Index Terms—Boundary conditions, coronary artery disease, fractional flow reserve, image segmentation, reduced-order modeling, uncertainty.

I. INTRODUCTION

FRACTIONAL flow reserve (FFR) is the pressure ratio that decides whether a coronary narrowing is treated. It is the blood pressure beyond the narrowing divided by the aortic pressure during maximal flow. A value at or below 0.80 marks a stenosis for revascularization [1]. FFR can also be computed without a pressure wire. The lumen, the blood-filled channel of the artery, is segmented from coronary computed tomography (CT) angiography and converted into

a flow model, and the computed FFR has been validated against invasive measurement in multicenter trials [2], [3]. The calculation exists in three-dimensional (3D), reduced-order and machine-learning forms [4], [5], [6].

Every such model needs boundary conditions, the pressures, flows or resistances imposed where the model ends to stand in for the vessels the image cannot resolve. The largest part of this unresolved circulation is the microvascular bed, the arterioles and capillaries downstream of the segmented arteries, which set how much blood each region of heart muscle draws. Its resistance is not measured but derived from the segmented lumen through scaling laws that relate vessel size to the flow it carries [7], [4]. It is increasingly tuned so that the model reproduces measured perfusion, the blood flow to each region of heart muscle, or measured flow splits [8], [9]. The segmentation therefore enters the computed FFR twice, through the resistance of the arteries it describes and through the boundary conditions derived from them.

Segmentation error has been studied mainly as caliber error. Caliber is the inner diameter of a vessel, so a caliber error is a lumen drawn too wide or too narrow, or a narrowing drawn too long, while the branching is unchanged. Sensitivity analyses perturb the radius at fixed bifurcations and report a continuous change in the computed pressure index [10], [11], [12], [13]. Studies that set geometry against boundary conditions rank the two by their influence on a continuous output [14], [15], [16]. Topological error, a change in the branching structure itself such as a missed side branch or a vessel that ends too early, has received less attention. In a pulmonary network model, connectivity contributed more to hemodynamic uncertainty than vessel radius and length [17], and examples from a recent coronary benchmark showed vessel breaks in the output of every evaluated method despite high overlap scores [18], and small distal segments remain the hardest to segment [19]. A missed branch removes part of the microvascular bed from the model and changes both the flow through the stenosis and the boundary conditions derived from the tree.

Whether such an error reaches the computed FFR depends on how the boundary conditions are handled. With outlet resistance held fixed, adding a side branch beyond a stenosis lowered FFR by 15.5% in an idealized model and by 13% in one of two models built from optical coherence tomography [20]. When a cohort-wide outlet resistance was recalibrated for a model with side-branch outflow, diagnostic accuracy changed little [21]. In a clinical cohort, a change of outlet model left the area under the receiver operating characteristic

F. Yamin, X. Wang and M.-Z. P. Ismadi are with the Department of Mechanical Engineering, School of Engineering, Monash University Malaysia, 47500 Bandar Sunway, Selangor, Malaysia (corresponding author: M.-Z. P. Ismadi, e-mail: mohd.zulhilmi@monash.edu).

M. A. Mohammed Sapardi is with the Department of Mechanical Engineering, Kuliyah of Engineering, International Islamic University Malaysia, 53100 Kuala Lumpur, Malaysia.

M. K. Tan is with the Department of Mechanical Engineering, School of Engineering, Monash University Malaysia, and the Department of Mechanical Engineering, Ming Chi University of Technology, New Taipei City 243, Taiwan.

curve unchanged at 0.845 while sensitivity moved from 58.1% to 68.6% [5]. If the boundary conditions are tuned until the model matches measured perfusion, a topological error may vanish from the validation check yet remain in the pressure along the vessel. None of these studies compares fixed, re-derived and tuned boundary conditions on the same corrupted anatomy at 0.80, or tests whether a model can pass a perfusion check and still assign the wrong treatment.

We propose a controlled test for this problem, an ablation in which one segmentation error at a time is introduced into an otherwise correct model. The correct (clean) model of each tree serves as the reference, so its FFR is the true value and every change in FFR can be traced to the error and to the handling of the boundary conditions. We inserted 150 stenoses into 108 real coronary trees from a public CT angiography dataset so that the baseline FFR spans the decision threshold, applied two topological and two caliber error types, and recomputed FFR with a reduced-order model under fixed, re-derived and tuned boundary conditions in two microvascular bed structures. The approach is new in three respects. First, segmentation error is judged by whether it changes the treatment decision at 0.80, not by a continuous sensitivity, and against the variability of repeat invasive FFR. Second, the same corrupted anatomy is solved under fixed, re-derived and tuned boundary conditions, which isolates the effect of tuning, and topological errors are separated from caliber errors of measured inter-observer magnitude. Third, the test counts how often a model passes a perfusion check while its FFR is materially wrong, a validation failure that, to our knowledge, has not been quantified, and a 3D computational fluid dynamics (CFD) case study confirms the mechanism. The results show what a perfusion check misses and which further checks a computed-FFR pipeline needs.

II. METHODS

In outline, we inserted a stenosis into each real coronary tree, introduced one segmentation error at a time (Fig. 1), recomputed FFR under three boundary-condition protocols and two bed structures, and compared each result with the clean model of the same tree.

A. Data and Cohort

ImageCAS-X provides voxel-wise lumen masks, labeled centerlines and surfaces for 800 CT angiography scans of the ImageCAS dataset [22], [18], [23]. We used its 160-scan test split, in which a second analyst re-annotated each scan. The two annotators agreed to a Dice similarity coefficient (DSC, the overlap of two masks) of 92.8% and a 95th-percentile Hausdorff distance (HD95, the surface distance exceeded by 5% of boundary points) of 2.46 mm [18]. Each scan gives a left and a right coronary tree. The radius at each centerline point is the distance from the point to the lumen boundary in the mask, less half a voxel.

The dataset contains few lesions near the decision threshold, so we inserted one stenosis per instance into an otherwise unchanged tree. An instance is one lesion in one tree. A host vessel (left anterior descending, LAD; left circumflex, LCx; or

right coronary artery, RCA) was eligible when image quality was at least adequate (2 on a 0–4 scale), the healthy reference radius r_{fit} at the lesion center (from a linear taper fit) was at least 1.0 mm, native narrowing was below 40% diameter stenosis (DS), the measurement point 20 mm beyond the lesion lay on a vessel of radius at least 0.75 mm, and the FFR before insertion was at least 0.90. The lesion is an axisymmetric cosine narrowing (Supplementary Material),

$$r'(s) = \min(r, (1-w)r + w r_{\text{fit}}(1-DS)), \quad (1)$$

$$w = \frac{1}{2} \left[1 + \cos \frac{2\pi(s-c)}{L} \right]$$

for $|s-c| < L/2$, where s is arc length, c the lesion center and L its length. The vessel was never widened. Lesions were centered 20 or 45 mm from the vessel origin, were 10 or 20 mm long and had 40, 50, 55, 60, 65, 70, 75 or 80% DS. Of the 6944 resulting instances (280 host vessels, 140 scans), we drew 25 in each of six 0.05-wide bands of baseline FFR from 0.65 to 0.95, at most two per tree, giving 150 instances (50 per host vessel) in 108 trees from 93 patients.

B. Reduced-Order FFR Model

All 150 instances were solved with a reduced-order model, which represents the coronary tree as a network of flow resistances instead of computing the full 3D flow field. The model places one node at every centerline point. Each element between consecutive nodes has the Poiseuille resistance $8\mu\Delta s/(\pi r^4)$ of its actual radius r and length Δs , with blood viscosity $\mu = 0.004$ Pa·s. Where the narrowing exceeds 30% DS, an expansion loss $\Delta p = K Q|Q|$ is added at the throat, with $K = \rho K_t (A_0/A_s - 1)^2 / (2A_0^2)$, $K_t = 1.52$ [24], density $\rho = 1060$ kg m⁻³, and A_0 and A_s the reference and throat areas. Aortic pressure is 90 mmHg and venous pressure 5 mmHg. Because the throat loss depends on flow, the equations are nonlinear and have no analytical solution for a real tree. They are solved numerically by under-relaxed fixed-point iteration, each step a sparse linear solve for the node pressures, until the relative flow change falls below 10^{-8} . The mass-conservation error is recorded for every solve. FFR is the computed pressure at the measurement point divided by aortic pressure. The model is steady and represents maximal hyperemia (maximal flow) without autoregulation.

The microvascular bed is modeled as conductances (the inverse of resistance) that drain flow from network nodes to venous pressure. Each conductance depends on a healthy reference radius r_{ref} , the taper-fit radius r_{fit} made non-increasing along each path, and is scaled by one bed constant C . Two bed structures were analyzed because they treat a deleted vessel differently. In the leaky bed, flow leaves along the vessel wall at every node of the tree truncated at $r_{\text{ref}} = 0.50$ mm, with conductance $(r_{\text{ref}}^3 - \sum r_{\text{ref,child}}^3)/C$, so that flow follows Murray's cube law [7]. This distributed outflow stands for the side branches the image does not resolve, and the conductance of a deleted branch reappears as wall outflow at its parent node. In the discrete bed, flow leaves only at the outlets of the tree truncated at $r_{\text{ref}} = 0.60$ mm, with conductance $r_{\text{ref}}^{2.66}/C$, where the exponent approximates 8/3, the inverse of the 3/8 power that relates coronary vessel diameter to the

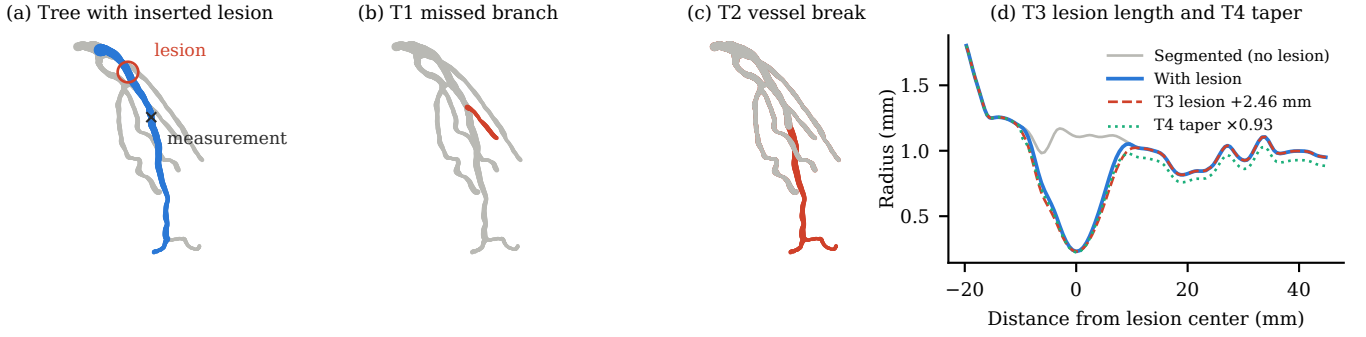


Fig. 1. The four segmentation error types on one cohort tree (left coronary tree of scan 14, front view, line width proportional to radius). (a) The tree with an inserted 80% DS lesion in the proximal LAD (blue) and the measurement point 20 mm beyond it. (b) T1 deletes the largest side branch beyond the lesion (red). (c) T2 ends the host vessel 25 mm beyond the lesion, and the red part is lost. (d) Radius along the LAD for the caliber errors, T3 lengthening the lesion by 2.46 mm and T4 shrinking the radius by 7% from the lesion onwards. The same instance was solved in 3D.

myocardial mass it perfuses [25], so that outlet flow is taken as proportional to perfused mass. The discrete bed is the outlet structure that a 3D model can represent. In both structures C was calibrated on the healthy-equivalent network, the tree with its reference radius and no lesion, so that inflow equals the hyperemic demand $Q = k r_{in}^3$, where r_{in} is the inlet radius and the cube law is Murray's [7]. With the model setting $k = 562 \text{ s}^{-1}$, a vessel with the normal proximal LAD diameter of 3.7 mm [26], carries 214 mL/min, within one standard deviation of the hyperemic flow measured in that artery by continuous thermodilution (228 ± 71 to 293 ± 102 mL/min) [27]. The cohort's median demand was 1.5 mL/s (inlet radius 1.39 mm). Reference radii and C are computed from the original tree, so an inserted lesion changes only the epicardial radius. The leaky bed was analyzed on all 150 instances. The discrete bed was analyzed on the 97 instances whose healthy-equivalent network gave a main-vessel FFR of at least 0.90 and had at least two outlets, and baseline FFR bands were derived separately for each bed. Halving the centerline spacing changed FFR by at most 0.005 in either bed.

C. Segmentation Error Types

Four error types were applied to each instance with its lesion in place. Two are topological errors, which change which vessels exist. Two are caliber errors, which change the lumen size or lesion extent while the branching is unchanged. The missed branch (T1) deletes the largest side branch that leaves the host vessel beyond the lesion, together with its subtree. The vessel break (T2) ends the host vessel 25 mm beyond the distal edge of the lesion, so the measurement point survives but the run-off beyond it is lost. The lesion length (T3) re-inserts the same stenosis 2.46 mm longer. The taper (T4) multiplies the radius by 0.930 from the proximal edge of the lesion onwards and in every descendant vessel. For T3 and T4 the set of modeled nodes is fixed to that of the clean tree, so a narrowed radius does not remove an outlet.

The caliber magnitudes are the inter-observer statistics of the test split, so each caliber error is as large as the disagreement between two annotators. HD95 (2.46 mm) sets T3. The DSC of 92.8% sets T4 as the radius ratio k of two coaxial cylinders, $2k^2/(k^2 + 1) = 0.928$. Because both

annotators edited the same automatically generated centerlines, the dataset authors describe the inter-observer DSC as an upper bound on agreement [18]. T1 and T2 have no measured magnitude and are design choices. Nodes of the corrupted tree were matched to the clean tree by coordinate, and FFR was read at the same anatomical point.

D. Boundary-Condition Protocols

Each corrupted tree was solved under three protocols. Under Protocol A (fixed), surviving nodes keep the bed conductances and C of the clean model, and the bed of deleted vessels is lost. This represents the case in which the boundary conditions were obtained correctly despite the segmentation error. Under Protocol B (re-derived), the bed rule is applied to the corrupted tree and C is recalibrated to that tree's own hyperemic demand, as an automated pipeline does with whatever anatomy it receives. Under Protocol C (tuned), one global scaling of C (one bed scaling) is fitted so that the corrupted model reproduces the clean model's territory flows. A perfusion territory is the subtree below each child of the first bifurcation of the clean tree. Its target is the total bed outflow of that subtree in the clean model, including the share of any vessel the error deleted, because the heart muscle is perfused whether or not its artery was segmented. Each corrupted node belongs to the territory of its clean counterpart. The fit minimized the mean squared relative error of the territory flows over a 61-point grid spanning ± 1.5 decades of C about the re-derived value, followed by bounded refinement. A fit at the edge of the search range was recorded as a failure. One parameter against two or more targets makes the fit over-determined, so this is the least flexible tuning a pipeline could use. Protocol C is therefore defined only when at least two territories retain a vessel. It was undefined for 33 of 77 (discrete) and 47 of 118 (leaky) missed-branch instances and for 36 of 96 and 49 of 149 vessel breaks, all but four in the RCA (a missed branch is possible only where a side branch leaves the host vessel beyond the lesion). It matches flow only, because matching both flow and pressure at the outlets of a steady tree would reproduce the clean FFR by construction.

For every protocol we computed the perfusion residual, the root mean square of the relative differences between the

model's territory flows and the clean targets. It is the quantity a perfusion check compares with the measurement.

E. Three-Dimensional Case Study

To test whether the findings depend on the simplifications of the reduced-order model, one instance (a 20 mm lesion of 80% DS in the proximal LAD) was also solved with 3D CFD, which computes the full velocity and pressure fields from the Navier–Stokes equations. These have no analytical solution for a real artery, so they were solved numerically by the finite-volume method. Three lumens were built: clean, with the lesion (baseline), and with the lesion and the missed branch (T1). Surfaces were extracted from the mask by marching cubes and Taubin smoothing; the lesion was applied by moving surface vertices radially with (1), and the branch was removed by deleting its voxels. Meshes of 3.5–3.9 million cells (cfMesh) had 25 μm cells within ± 4 mm of the throat and four boundary layers.

Steady laminar Newtonian flow, with the viscosity and density of the reduced-order model, was solved with the steady incompressible solver of OpenFOAM (simpleFoam, ESI v2406, SIMPLE pressure–velocity coupling), with a bounded second-order upwind scheme for convection and second-order central schemes elsewhere. Each solve ran 3 000 iterations; convergence required scaled residuals below 10^{-5} and a steady pressure at the measurement point. The walls were rigid with no slip, and the inlet carried aortic pressure. Each outlet either set its pressure to venous pressure plus the clean tree's resistance times its flow (Protocol A), or received a prescribed clean territory flow, shared over the territory's surviving outlets by bed weight (the per-outlet form of Protocol C). FFR is the area-averaged pressure on a cross-section at the measurement point divided by aortic pressure. A grid-convergence study on an idealized 70% DS stenosis (throat cells of 50, 25 and 12.5 μm) gave a discretization uncertainty in FFR of 5.5×10^{-4} . A reduced-order twin of each geometry, rebuilt on the area-equivalent radius of the meshed lumen with the same outlet conditions, separates the effect of model fidelity from that of radius definition.

F. Outcomes and Statistical Analysis

For each instance, error type, protocol and bed, ΔFFR is the corrupted model's FFR minus that of the instance's own clean model in the same bed. A flip is a change of classification at 0.80 between the two, that is, a change in the treatment decision. A model passes the perfusion check when its perfusion residual is below 10%. Published repeatability of hyperemic myocardial blood flow (within-subject variation of 10–15% [28] and a regional repeatability coefficient of 30–37% [29]) implies a 95% bound of about 20–29% for one model output against one regional measurement, so the 10% threshold is strict, and a looser one can only increase the number of models that pass. Thresholds of 13% and 16% were analyzed as sensitivities. An FFR is materially wrong when $|\Delta\text{FFR}| > 0.05$, an error large enough to matter near the 0.80 cut.

Proportions are reported with Wilson 95% confidence intervals (CIs). Protocol contrasts (B against A, C against A, C against B) are paired within instance and tested with the exact McNemar test, with Holm adjustment across the three contrasts of each error type and bed. Topological and caliber errors are compared by a paired sign test on the per-instance flip proportions of each class, and paired changes in $|\Delta\text{FFR}|$ by the Wilcoxon signed-rank test. A Bayesian mixed-effects logistic model of flips with patient and lesion-slot random effects is reported in the Supplementary Material. Every analysis was run in both beds, effect sizes are reported per bed, and a finding is claimed only where its direction agrees in both.

Each flip rate is compared with a noise floor, the rate expected from measurement variability alone. For each instance, this is the probability that a repeat invasive FFR, drawn from a normal distribution centered on the clean value with the published standard deviation of the difference between repeat measurements, 0.018 [30], falls on the other side of 0.80, following the measurement-certainty approach of [31]. A second floor was simulated on the clean anatomy: 20 draws per instance perturbed cardiac output, aortic pressure, blood viscosity, the share of flow to each territory and the measured territory flows by their within-subject variability, and the model was tuned to the perturbed flows as in Protocol C.

The cohort, design, analysis plan and ablation code were fixed, and the cohort list hashed, before the ablation was run. The analysis script was written while the ablation ran and before its output was read. The analysis pipeline is shown in the Supplementary Material.

III. RESULTS

A. Decision Changes by Error Type and Protocol

Topological errors changed the decision more often than caliber errors under every protocol and in both beds (Table I). With fixed boundary conditions in the discrete bed, 45 of 137 topological-error models (33%, 95% CI 26–41%) crossed 0.80, against 20 of 194 caliber-error models (10%, 7–15%). In the leaky bed the proportions were 32% (26–38%) and 6% (4–9%) (paired sign test, $p \leq 0.002$ in both beds). Under Protocols B and C the difference kept its direction. It was significant in the discrete bed ($p = 0.013$ and 0.002) but not in the leaky bed ($p = 0.18$ and 0.057). Flips were concentrated in the bands adjacent to 0.80 (Fig. 2). All 2 599 corrupted-model solves converged (maximum mass-conservation error 9×10^{-9}); in the discrete bed the vessel break removed every outlet in 36 instances under Protocol A.

The caliber errors rarely changed the decision. At the magnitude of inter-observer disagreement they changed FFR by a mean of 0.006–0.037 in absolute value, and their flip rates of 1–13% were of the order of the 5.0–6.5% expected from repeat invasive measurement. Only the taper under fixed boundary conditions exceeded this noise floor with its lower confidence bound, at 13% in the discrete bed and 9% in the leaky bed (Table I).

Re-deriving the boundary conditions from the corrupted tree (Protocol B) produced the fewest flips. Against Protocol A, it

TABLE I
DECISION CHANGES AND CONCEALMENT BY ERROR TYPE, PROTOCOL AND BED STRUCTURE

Error	Protocol	Discrete bed (97 instances)			Leaky bed (150 instances)		
		<i>n</i>	Flip, %	Passes and wrong, %	<i>n</i>	Flip, %	Passes and wrong, %
T1	A	77	27 (19–38)	13 (7–22)	118	18 (12–26)	16 (11–24)
	B	77	9 (4–18)	1 (0–7)	118	5 (2–11)	5 (2–11)
	C	44	18 (10–32)	20 (11–35)	71	8 (4–17)	11 (6–21)
T2	A	60	40 (29–53)	12 (6–22)	147	43 (35–51)	24 (17–33)
	B	96	18 (11–27)	2 (0–9)	149	5 (2–9)	4 (2–10)
	C	60	20 (12–32)	18 (11–30)	100	8 (4–15)	6 (3–12)
T3	A	97	7 (4–14)	0 (0–4)	150	2 (1–6)	0 (0–2)
	B	97	7 (4–14)	0 (0–4)	150	2 (1–6)	0 (0–2)
	C	97	7 (4–14)	0 (0–4)	150	3 (1–7)	1 (0–4)
T4	A	97	13 (8–22)	6 (3–13)	150	9 (6–15)	7 (4–12)
	B	97	8 (4–15)	0 (0–4)	150	1 (0–4)	0 (0–2)
	C	97	8 (4–15)	9 (5–17)	150	5 (3–10)	24 (18–31)

A flip is a change of classification at FFR 0.80. Passes and wrong means a territory-perfusion residual below 10% and $|\Delta\text{FFR}| > 0.05$, over models with a defined residual (T2 under B in the discrete bed, $n = 60$; T2 under A and B in the leaky bed, $n = 100$). Wilson 95% intervals in parentheses. The expected flip rate from repeat invasive measurement is 5.0–6.5% across cells.

reversed 14 of 21 missed-branch flips in the discrete bed and 15 of 21 in the leaky bed, and created none (McNemar, Holm-adjusted $p < 0.001$ for both). It reversed 56 of 63 vessel-break flips in the leaky bed ($p < 0.001$). In the discrete bed it reversed 22 vessel-break flips but created 12 ($p = 0.24$), because the re-derived bed lowered FFR below the clean value (mean $\Delta\text{FFR} -0.039$). Protocol B models, however, failed their own perfusion check most often.

B. Concealment Under Tuned Boundary Conditions

We call it concealment when tuning removes the error from the perfusion check but not from the FFR. Fitting one bed scaling to the clean territory flows (Protocol C) lowered the median perfusion residual of topological-error models from 0.19 to 0.14 in the discrete bed and from 0.10 to 0.02 in the leaky bed. Compared with Protocol A, tuning reversed 4 missed-branch flips and created none in each bed ($p = 0.25$), and for vessel breaks reversed 32 and created none (leaky, $p < 0.001$) or reversed 22 and created 10 (discrete, $p = 0.15$). We had expected tuning to leave the flip rate at or above that of fixed boundary conditions. Instead, where the change was resolved, it was a reduction. Relative to re-derived boundary conditions (Protocol B), tuning changed at most two topological-error decisions per cell ($p \geq 0.5$), while it lowered the median residual of missed-branch models from 0.36 to 0.16 (discrete) and from 0.04 to 0.03 (leaky).

Tuning did not remove the error from the FFR (Fig. 3). Of the tuned topological-error models, 20 of 104 (19%, 13–28%) in the discrete bed and 14 of 171 (8%, 5–13%) in the leaky bed passed the perfusion check while their FFR was materially wrong (more than 0.05 from the clean value). In the discrete bed these were half of the 39 models that passed. Four and 12 of the passing models, respectively, also changed the decision. Under Protocol A the proportions were 12% and 20%, and because the ordering of A and C differed between the beds, no difference between them is claimed. Models with correct anatomy, tuned to perfusion measured with physiological noise, passed while materially wrong in

3.1% (2.4–4.0%) and 3.9% (3.2–4.6%) of draws, and their flip rates were 6.2% and 7.2%. The proportion depended on the assumed hyperemic demand. With the demand scaled by 0.7 and 1.3, it was 12% and 25% in the discrete bed and 3% and 12% in the leaky bed, but the ordering of topological against caliber errors and of Protocols B and C against A was unchanged (Supplementary Material). At the looser validation thresholds of 13% and 16%, the proportion under Protocol C rose to 28% (20–37%) and 38% (30–48%) in the discrete bed and was 9% (6–15%) at both in the leaky bed.

Tuning also enlarged the taper error. In the leaky bed, 36 of 150 tuned taper models (24%, 18–31%) passed while materially wrong, against 10 under Protocol A (Table I).

C. Side-Branch Loss Under Fixed and Tuned Resistance

A missed side branch raised the computed FFR, and tuning removed only part of the rise. In the 44 (discrete) and 71 (leaky) instances in which Protocol C was defined, the missed branch raised FFR by a median of 0.095 and 0.047 with fixed boundary conditions. The shift exceeded 13% of the clean FFR in 45% and 11% of these instances. Tuning reduced the shift to a median of 0.077 and 0.015 (Wilcoxon signed-rank, $p < 0.001$ in both beds; Fig. 4). In the leaky bed the tuned shift was a median 42% of the fixed shift, and 12 of 71 instances remained above 0.05. In the discrete bed it was 84%, and 33 of 44 remained above 0.05.

D. Three-Dimensional Case Study

The 3D solution showed the same concealment. With the clean tree's resistances at the outlets, removal of the side branch raised the 3D FFR at the measurement point from 0.870 to 0.944 ($\Delta\text{FFR} = +0.074$). With the clean territory flows prescribed at the outlets, the same error changed FFR by -0.0007 (0.892 in both geometries). The reduced-order twins gave $+0.081$ and -0.0005 for the same two conditions, and their baseline FFR was within 0.011 of the 3D values (Fig. 5). Two caveats apply. The meshed lumen was wider than the radius used by the reduced-order model (median

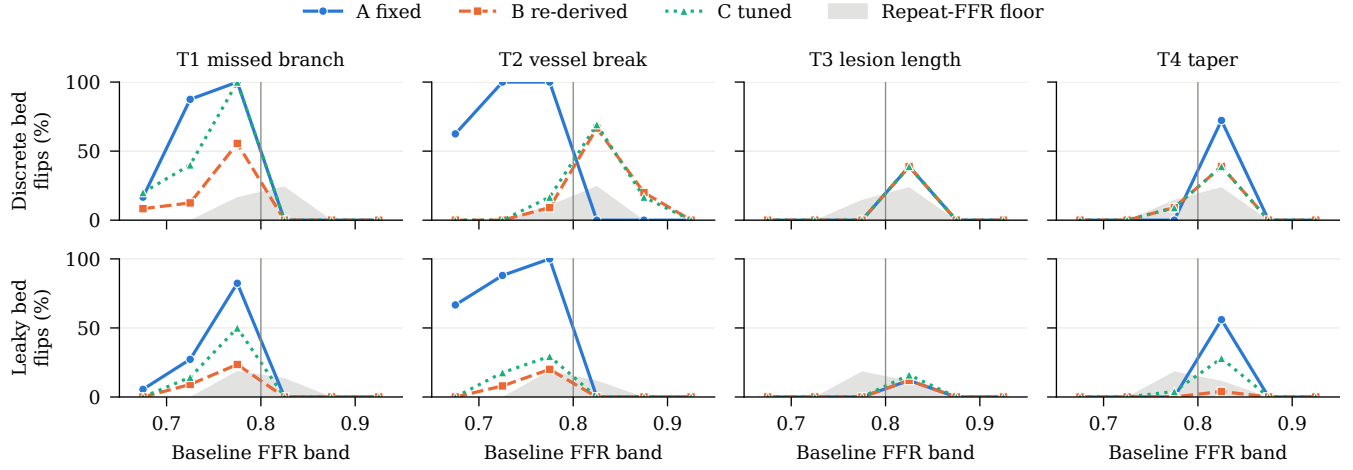


Fig. 2. Flip rate (proportion of models whose classification at FFR 0.80 changed) by baseline FFR band (0.05 wide, plotted at the band center; bands with fewer than three models omitted), error type, protocol and bed structure. The shaded area is the flip rate expected from repeat invasive FFR alone (standard deviation 0.018).

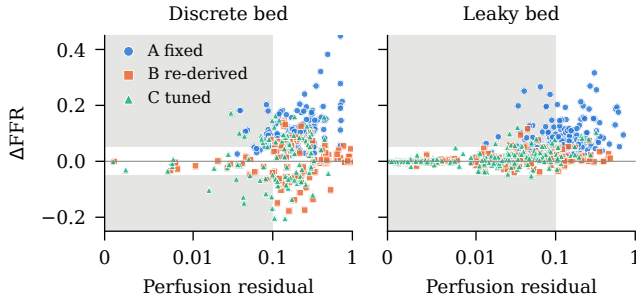


Fig. 3. Change in FFR against territory-perfusion residual for missed-branch and vessel-break models. Each point is one model. The shaded areas hold models that pass the perfusion check (residual < 0.10) while materially wrong ($|\Delta\text{FFR}| > 0.05$). The horizontal axis is linear below 0.01 and logarithmic above.

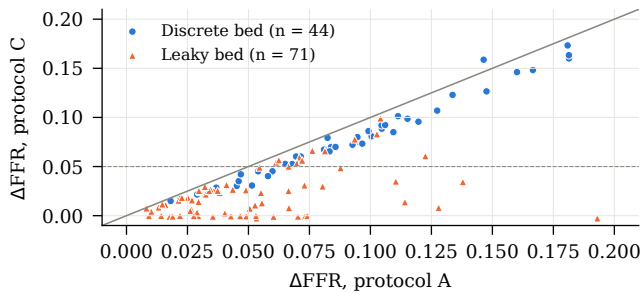


Fig. 4. Change in FFR caused by a missed side branch under fixed boundary conditions (Protocol A) and after tuning to territory flows (Protocol C), paired by instance. The solid line is the identity and the dashed line marks $\Delta\text{FFR} = 0.05$.

difference 0.14 mm; throat area-equivalent radius 0.276 against 0.231 mm), and on the requested radius the reduced-order baseline was 0.761. Halving the cell size in the throat zone to 12.5 μm changed the baseline FFR by 0.0019 and 0.0021 in two zone placements, against the error effect of 0.074, and

a regenerated mesh at 25 μm reproduced the baseline value (Supplementary Material).

IV. DISCUSSION

A. Concealment at Calibration

We asked whether tuning a coronary flow model to perfusion can hide a segmentation error that changes the treatment decision. Three findings answer this. First, the decision risk of segmentation lies mainly in the branching structure of the lumen, not in its caliber. With fixed boundary conditions, a topological error (a missed branch or a vessel break) changed the decision at 0.80 three to six times as often as a caliber error of inter-observer magnitude. The caliber errors produced flip rates of the order of repeat invasive measurement.

Second, how much of a topological error reaches the decision depends on the boundary conditions. Re-deriving or tuning the bed reduced topological flips from 18–43% to 5–20%. Tuning moved the decision toward the clean result, not away from it.

Third, tuning separated the validation check from the error. Re-derived models carried the error in their perfusion residual, so a perfusion check flagged them. Tuning removed most of that residual while leaving the decisions almost unchanged, and in the discrete bed half of the tuned topological-error models that passed the check were materially wrong. The check measured agreement with the fitted targets, not the correctness of the model. The remaining error lay in the pressure along the vessel, which one global scaling cannot restore once a branch is absent, and the three-dimensional case showed the same mechanism.

Bed structure set the magnitude. In the leaky bed a deleted branch's conductance reappears as wall outflow at its parent node, damping the error before tuning, whereas the discrete bed, which matches 3D outlets, kept 84% of the missed-branch shift after tuning. Claims are therefore limited to directions that held in both.

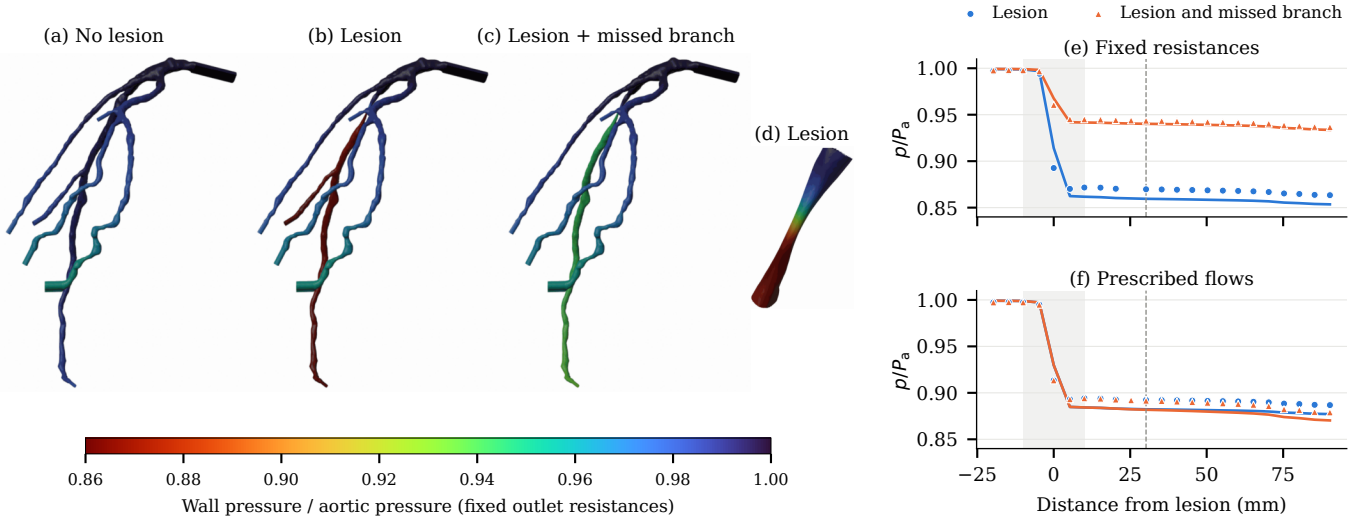


Fig. 5. Three-dimensional case study (proximal LAD, 80% DS). (a)–(c) Wall pressure relative to aortic pressure with the clean tree’s outlet resistances, without the lesion, with the lesion, and with the lesion and a missed side branch. The lesion lowers the downstream pressure (red); the missed branch raises it again (green), a shift that would move a decision near 0.80. (d) Close-up of the lesion. (e), (f) Pressure along the artery with fixed resistances and with prescribed clean territory flows. Markers show 3D cross-section averages and lines the reduced-order models on the meshed radius with the same outlet conditions. The shaded band is the lesion and the dashed line the measurement point.

B. Relation to Published Studies

Published sensitivity analyses of computed pressure indices perturb lumen caliber with the branching structure fixed and report continuous sensitivity [10], [11], [12], [13]. Studies that compare geometry with boundary conditions rank their influence on a continuous output [14], [15], [16]. Our results add that the ranking depends on the error type and the boundary-condition protocol, and that the decision outcome separates them more clearly.

Two earlier findings that appeared to disagree are reproduced here. A side-branch effect of 13–15.5% under fixed outlet resistance was reported in an idealized model and a model built from optical coherence tomography [20], whereas recalibrating a cohort-wide outlet resistance absorbed a large change in side-branch flow with little change in diagnostic accuracy [21]. Both appear in our cohort, the first with fixed resistance and the second after tuning, so the difference follows from the boundary-condition protocol. This is consistent with a clinical cohort in which a change of outlet model left the area under the curve unchanged while sensitivity changed [5].

Vessel breaks occur in the output of current segmentation methods despite high overlap scores [18], and a missing side branch removes few voxels, so the error types that carried the decision risk here are the ones that the usual overlap metrics do not measure, although topology-aware measures such as cIDice exist [32].

C. Implications and Practical Use

Perfusion imaging is used to personalize coronary flow models [8], but agreement with a fitted quantity does not by itself make a model credible, which also requires verification, uncertainty quantification and applicability [33]. Because a

missed branch raised FFR, its concealed error biases the decision toward deferring treatment of a lesion that needs it.

For a user who computes FFR from an image, the results suggest four steps. First, check that the side branches beyond the lesion are present and that no vessel ends early, because overlap scores do not reveal these errors. Second, treat diameter errors of inter-observer size as a minor risk, since they flipped decisions about as often as repeat invasive measurement. Third, if the boundary conditions are tuned to perfusion, record the mismatch before tuning, because a large mismatch marked a missing branch that tuning later hid. Fourth, take most care near 0.80, where decision changes concentrated. A developer can also apply the same test to a pipeline, introducing the four errors into its own cases and comparing the resulting flip and pass-while-wrong rates with Table I and the noise floors. These steps follow from the results but are not validated decision rules.

D. Limitations

The inserted stenoses are idealized, and the reduced-order model is steady, without autoregulation or heart-rate effects. The missed-branch and vessel-break magnitudes are design choices, and the caliber magnitudes come from inter-observer statistics that likely overstate agreement. Protocol C used one global scaling, the least flexible tuning, and was undefined for most right coronary instances with a topological error, so its results for these errors describe the left coronary tree. The 3D analysis covers one instance, with a meshed lumen wider than the reduced-order radius; this offset shifts absolute FFR but not the paired differences on which the findings rest. No invasive FFR was available, so the noise floor was modeled from published repeat measurements.

V. CONCLUSION

In 150 controlled stenoses in 108 coronary trees, with fixed boundary conditions, topological segmentation errors changed the decision at FFR 0.80 three to six times as often as caliber errors of inter-observer magnitude, which were close to the repeatability of invasive measurement. The boundary-condition protocol alone changed how often a topological error altered the decision, from 43% of models with fixed boundary conditions to 5% with re-derived ones for a vessel break in the leaky bed. Tuning to territory flows reduced decision changes but removed the perfusion residual that marked the error, and 8–19% of tuned models passed a perfusion check while materially wrong. A three-dimensional case reproduced this concealment.

DATA AND CODE AVAILABILITY

ImageCAS-X is available under CC BY 4.0 (doi:10.5281/zenodo.21887809). The code and the frozen cohort list are available from the corresponding author on request.

ACKNOWLEDGMENT

This work received no external funding, and the authors declare no conflicts of interest. The study used a public, anonymized dataset and required no ethics approval.

REFERENCES

- [1] P. A. L. Tonino *et al.*, “Fractional flow reserve versus angiography for guiding percutaneous coronary intervention,” *N. Engl. J. Med.*, vol. 360, no. 3, pp. 213–224, 2009.
- [2] C. A. Taylor, T. A. Fonte, and J. K. Min, “Computational fluid dynamics applied to cardiac computed tomography for noninvasive quantification of fractional flow reserve: Scientific basis,” *J. Amer. Coll. Cardiol.*, vol. 61, no. 22, pp. 2233–2241, 2013.
- [3] B. L. Nørgaard *et al.*, “Diagnostic performance of noninvasive fractional flow reserve derived from coronary computed tomography angiography in suspected coronary artery disease: The NXT trial (Analysis of Coronary Blood Flow Using CT Angiography: Next Steps),” *J. Amer. Coll. Cardiol.*, vol. 63, no. 12, pp. 1145–1155, 2014.
- [4] H. J. Kim, I. E. Vignon-Clementel, J. S. Coogan, C. A. Figueroa, K. E. Jansen, and C. A. Taylor, “Patient-specific modeling of blood flow and pressure in human coronary arteries,” *Ann. Biomed. Eng.*, vol. 38, no. 10, pp. 3195–3209, 2010.
- [5] F. E. Fossan *et al.*, “Relevance of autoregulation in the diagnosis of coronary artery disease with CT-FFR,” *Amer. J. Physiol. Heart Circ. Physiol.*, vol. 330, no. 1, pp. H157–H169, 2026.
- [6] L. Itu *et al.*, “A machine-learning approach for computation of fractional flow reserve from coronary computed tomography,” *J. Appl. Physiol.*, vol. 121, no. 1, pp. 42–52, 2016.
- [7] C. D. Murray, “The physiological principle of minimum work: I. the vascular system and the cost of blood volume,” *Proc. Nat. Acad. Sci. USA*, vol. 12, no. 3, pp. 207–214, 1926.
- [8] K. Menon *et al.*, “Personalized coronary and myocardial blood flow models incorporating CT perfusion imaging and synthetic vascular trees,” *npj Imag.*, vol. 2, no. 1, 2024, art. no. 9.
- [9] M. Arminio *et al.*, “Patient-specific biomechanical investigation of percutaneous pulmonary valves: Towards the integration of routinely acquired clinical data and fluid-structure interaction simulations,” *IEEE J. Biomed. Health Inform.*, 2026, early access, doi: 10.1109/JBHI.2026.3708991.
- [10] S. Sankaran, L. Grady, and C. A. Taylor, “Fast computation of hemodynamic sensitivity to lumen segmentation uncertainty,” *IEEE Trans. Med. Imag.*, vol. 34, no. 12, pp. 2562–2571, 2015.
- [11] S. Sankaran, L. Grady, and C. A. Taylor, “Impact of geometric uncertainty on hemodynamic simulations using machine learning,” *Comput. Methods Appl. Mech. Eng.*, vol. 297, pp. 167–190, 2015.
- [12] D. Fernández-Martínez, M. R. González-Fernández, J. M. Nogales-Asensio, and C. Ferrera, “Impact of minimal lumen segmentation uncertainty on patient-specific coronary simulations: A look at FFR_{CT},” *Int. J. Numer. Methods Biomed. Eng.*, vol. 40, no. 6, 2024, art. no. e3822.
- [13] C. Dalmaso, F. E. Fossan, A. T. Bråten, and L. O. Müller, “Uncertainty quantification and sensitivity analysis for non-invasive model-based instantaneous wave-free ratio prediction,” *Int. J. Numer. Methods Biomed. Eng.*, vol. 41, no. 1, 2025, art. no. e3898.
- [14] S. Sankaran, H. J. Kim, G. Choi, and C. A. Taylor, “Uncertainty quantification in coronary blood flow simulations: Impact of geometry, boundary conditions and blood viscosity,” *J. Biomech.*, vol. 49, no. 12, pp. 2540–2547, 2016.
- [15] C. Tanade, S. J. Chen, J. A. Leopold, and A. Randles, “Analysis identifying minimal governing parameters for clinically accurate in silico fractional flow reserve,” *Front. Med. Technol.*, vol. 4, 2022, art. no. 1034801.
- [16] J. Korte *et al.*, “Is accurate lumen segmentation more important than outlet boundary condition in image-based blood flow simulations for intracranial aneurysms?” *Cardiovasc. Eng. Technol.*, vol. 14, no. 5, pp. 617–630, 2023.
- [17] M. J. Colebank *et al.*, “Influence of image segmentation on one-dimensional fluid dynamics predictions in the mouse pulmonary arteries,” *J. Roy. Soc. Interface*, vol. 16, no. 159, 2019, art. no. 20190284.
- [18] K. M. Bransby *et al.*, “ImageCAS-X: a dataset and benchmark for coronary artery segmentation and centerline extraction in coronary CT angiography,” arXiv:2608.30404, 2026.
- [19] F. Chen, J. Ge, Y. Zheng, K. Guo, F. Cao, and J. Liang, “Ensembled-SAMs for enhanced small coronary artery segmentation in CCTA images,” *IEEE J. Biomed. Health Inform.*, vol. 29, no. 7, pp. 4612–4622, 2025.
- [20] P. T. Gamage *et al.*, “Fractional flow reserve (FFR) estimation from OCT-based CFD simulations: Role of side branches,” *Appl. Sci.*, vol. 12, no. 11, 2022, art. no. 5573.
- [21] R. C. Gosling *et al.*, “Effect of side branch flow upon physiological indices in coronary artery disease,” *J. Biomech.*, vol. 103, 2020, art. no. 109698.
- [22] A. Zeng *et al.*, “ImageCAS: A large-scale dataset and benchmark for coronary artery segmentation based on computed tomography angiography images,” *Comput. Med. Imag. Graph.*, vol. 109, 2023, art. no. 102287.
- [23] K. M. Bransby and R. R. Paulsen, “ImageCAS-X,” Zenodo, 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.21887809>
- [24] D. F. Young and F. Y. Tsai, “Flow characteristics in models of arterial stenoses—I. Steady flow,” *J. Biomech.*, vol. 6, no. 4, pp. 395–410, 1973.
- [25] J. S. Choy and G. S. Kassab, “Scaling of myocardial mass to flow and morphometry of coronary arteries,” *J. Appl. Physiol.*, vol. 104, no. 5, pp. 1281–1286, 2008.
- [26] J. T. Dodge, B. G. Brown, E. L. Bolson, and H. T. Dodge, “Lumen diameter of normal human coronary arteries. Influence of age, sex, anatomic variation, and left ventricular hypertrophy or dilation,” *Circulation*, vol. 86, no. 1, pp. 232–246, 1992.
- [27] S. Fournier *et al.*, “Normal values of thermodilution-derived absolute coronary blood flow and microvascular resistance in humans,” *EuroIntervention*, vol. 17, no. 4, pp. e309–e316, 2021.
- [28] T. H. Schindler *et al.*, “Myocardial perfusion PET for the detection and reporting of coronary microvascular dysfunction: A JACC: Cardiovascular Imaging expert panel statement,” *JACC: Cardiovasc. Imag.*, vol. 16, no. 4, pp. 536–548, 2023.
- [29] L. A. E. Brown *et al.*, “Fully automated, inline quantification of myocardial blood flow with cardiovascular magnetic resonance: repeatability of measurements in healthy subjects,” *J. Cardiovasc. Magn. Reson.*, vol. 20, no. 1, 2018, art. no. 48.
- [30] N. P. Johnson *et al.*, “Repeatability of fractional flow reserve despite variations in systemic and coronary hemodynamics,” *JACC: Cardiovasc. Interv.*, vol. 8, no. 8, pp. 1018–1027, 2015.
- [31] R. Petraco *et al.*, “Fractional flow reserve-guided revascularization: Practical implications of a diagnostic gray zone and measurement variability on clinical decisions,” *JACC: Cardiovasc. Interv.*, vol. 6, no. 3, pp. 222–225, 2013.
- [32] S. Shit *et al.*, “cDice – A novel topology-preserving loss function for tubular structure segmentation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 16 555–16 564.
- [33] M. Viceconti *et al.*, “Position paper: Extending credibility assessment of in silico medicine predictors to machine learning predictors,” *IEEE J. Biomed. Health Inform.*, vol. 29, no. 7, pp. 5284–5290, 2025.